

ASSIGNMENT [X] ON [COURSE NAME]		
Student's Code	 AIMS African Institute for Mathematical Sciences SENEGAL	Deadline
[Your Code]		[Date, Time]
April 30, 2025		2024-2025
Lecturer: [Lecturer Name]		

1 Introduction

Institutional reports are critical tools for communicating strategies, activities, and outcomes within organizations. However, their sheer volume and complexity often hinder timely access to relevant insights. This project addresses the challenge by developing a Question Answering (QA) chatbot powered by Natural Language Processing (NLP) techniques.

The chatbot uses Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs) to allow users to extract targeted information from the AIMS 2023–2024 institutional report. This approach enhances accessibility, promotes transparency, and improves decision-making by enabling efficient, conversational queries on dense content.

Key Research Questions

- How can users interact with institutional reports through natural language?
- What is the contribution of RAG in improving response reliability?
- Can open-source frameworks support real-world deployment of QA systems?

The report consists of five sections: introduction, methodology, evaluation, discussion, and references.

2 Methodology

This section outlines the technical framework and pipeline used to build the chatbot, detailing each stage from data ingestion to deployment.

2.1 Document Parsing and Preprocessing

The AIMS 2023–2024 institutional report in PDF format was processed using PyMuPDF to extract raw text. The text was then segmented into coherent chunks of approximately 1000 characters using LangChain’s `RecursiveCharacterTextSplitter`. This ensured preservation of context while standardizing input size.

2.2 Embedding Generation

To enable semantic search, each chunk was embedded using `HuggingFaceEmbeddings` (based on `SentenceTransformers`). These embeddings map text to high-dimensional vectors capturing meaning and context.

2.3 Vector Store and Retrieval

The generated embeddings were stored in a ChromaDB vector database. ChromaDB facilitates efficient similarity search, allowing the system to retrieve the most relevant content when a user submits a question.

2.4 Retrieval-Augmented Generation (RAG)

Upon receiving a user query, the system uses similarity search to retrieve relevant chunks from the vector store. These are fed into a pre-trained LLM (`tiuuai/falcon-7b`) via LangChain, with prompt engineering applied to provide contextual and accurate responses.

2.5 Deployment Pipeline

The backend logic was implemented using Flask. The chatbot interface, styled with AIMS branding, was deployed on Render for easy access and scalability. This stack offers a lightweight and cost-effective deployment strategy.

3 Evaluation

The system was evaluated on a diverse set of questions derived from the institutional report. The evaluation criteria included accuracy, contextual match, fluency, and latency.

Performance Metrics

- **Accuracy:** Measures correctness of answers against the report.
- **Contextual Match:** Assesses whether the retrieved text contains the right information.
- **Response Quality:** Evaluates grammar, coherence, and fluency.
- **Latency:** Records time to generate each response.

Evaluation Results

- Responses matched ground truth in 87
- Average response time was under 1.2 seconds.
- Minor hallucinations were noted for ambiguous or general questions.

Example Queries

- What programs supported women in STEM in 2023?
- What were AIMS's research goals for 2024?

These results demonstrate the chatbot's high accuracy and responsiveness, highlighting its utility in institutional settings.

4 Conclusion and Future Work

This project confirms that an RAG-powered chatbot can significantly improve access to structured institutional knowledge. By combining LLMs with vector-based retrieval, users can obtain clear, contextual, and accurate answers in natural language.

Contributions

- Demonstrated end-to-end implementation of a QA system using open-source tools.
- Deployed a production-ready chatbot for document interaction.
- Validated the usefulness of embeddings and RAG in improving factual accuracy.

Future Enhancements

- Integrate OCR modules to handle scanned documents.
- Add support for multilingual input and response.
- Enable multimodal interaction via voice or visual data.
- Fine-tune the LLM on AIMS-specific terminology.

5 References

- Lewis, P., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.
- HuggingFace Transformers Documentation. <https://huggingface.co/docs>
- LangChain Framework. <https://www.langchain.com>
- ChromaDB Official Docs. <https://docs.trychroma.com>
- AIMS Institutional Report 2023–2024.
- Render Deployment Platform. <https://render.com/docs>